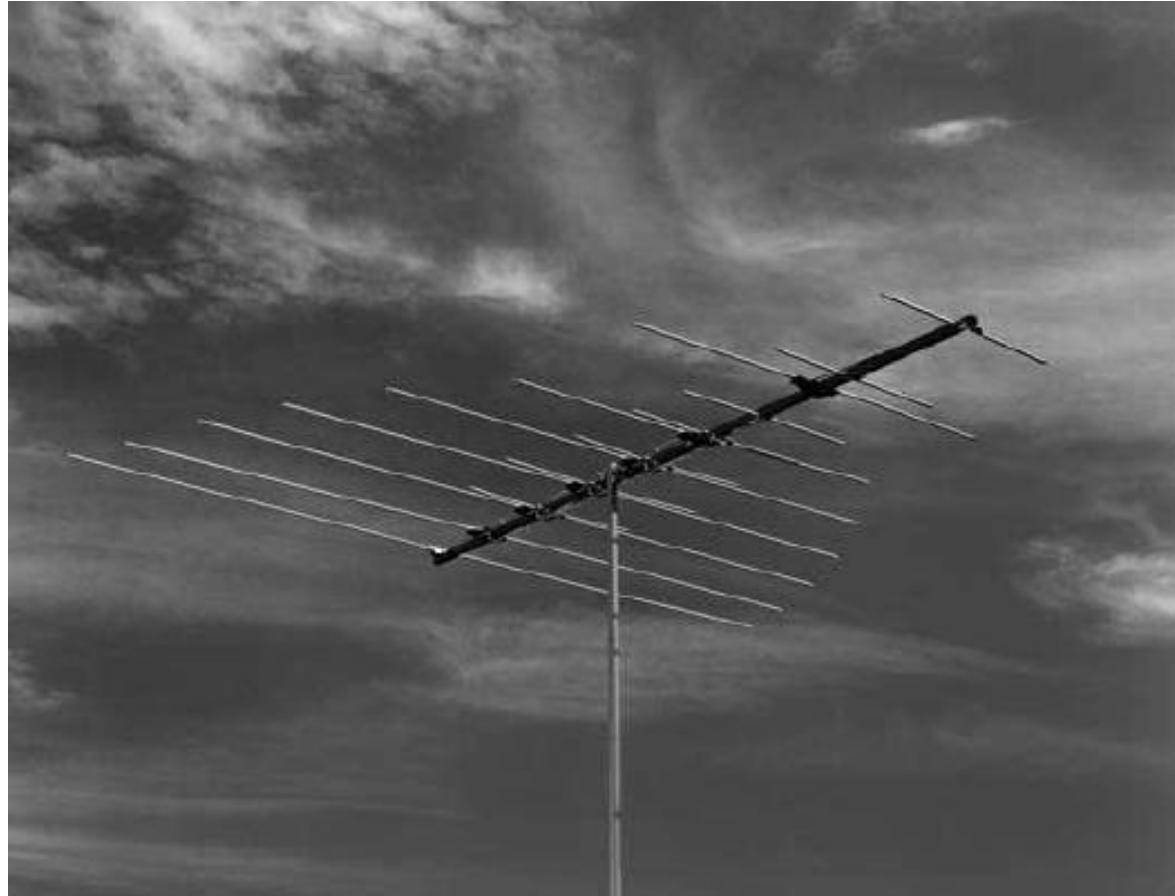


Polarization

Practical application



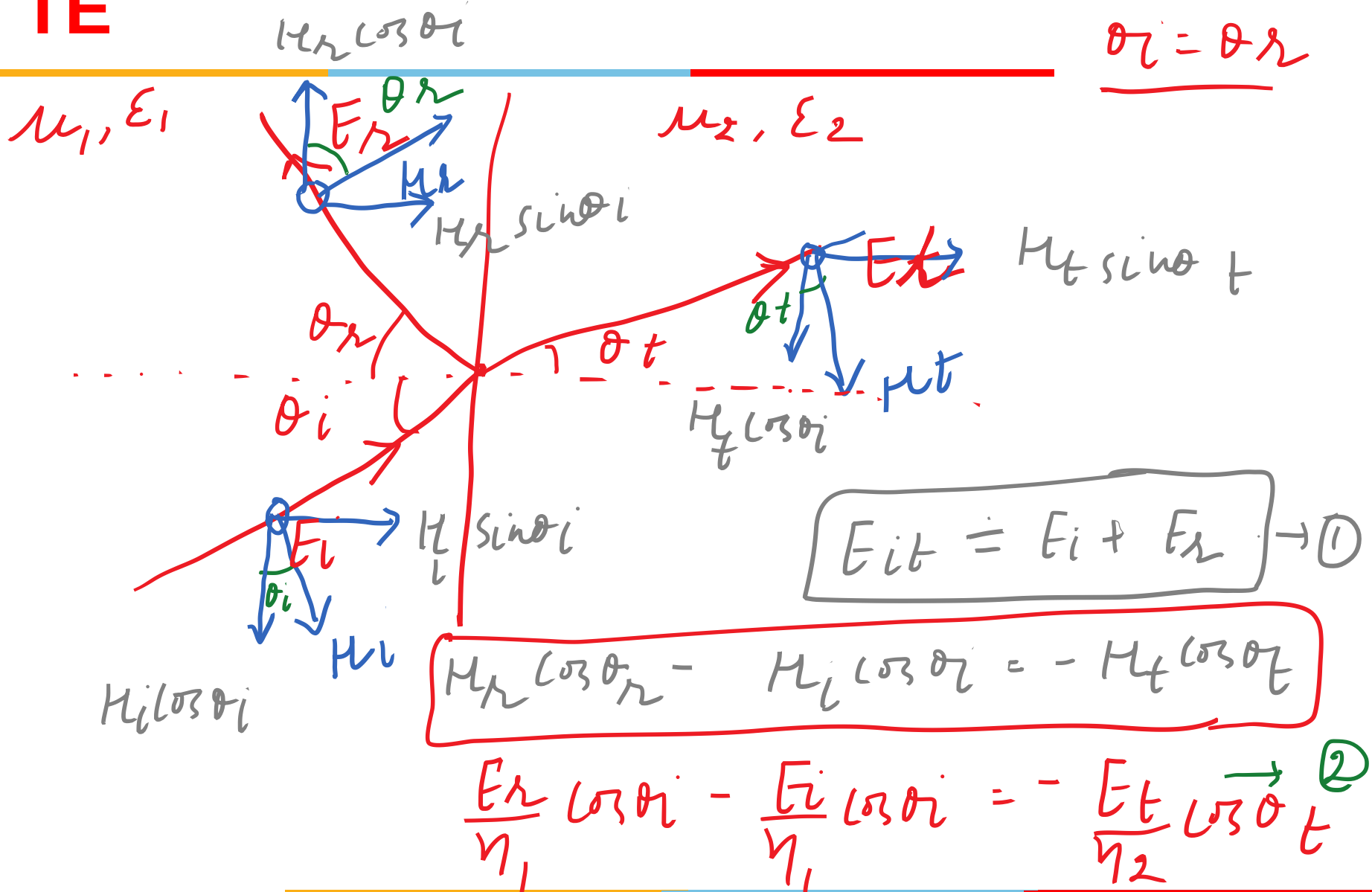
AM broadcast antenna



Roof top TV receiving antenna

Perpendicular polarization:

TE



Perpendicular polarization:

TE



Solving eqⁿ ① & ②

$$\left[\begin{aligned} \frac{E_r}{E_i} &= \frac{\eta_2 \cos \theta_i - \eta_1 \cos \theta_t}{\eta_2 \cos \theta_i + \eta_1 \cos \theta_t} \\ \frac{E_t}{E_i} &= \frac{2\eta_2 \cos \theta_i}{\eta_2 \cos \theta_i + \eta_1 \cos \theta_t} = \tau \end{aligned} \right]$$

Reflection coefficient

Transmission coefficient

$$E_{it} = E_i + E_r \rightarrow ①$$

$$H_r \cos \theta_r - H_i \cos \theta_i = -H_t \cos \theta_t$$

$$1 + \Gamma = \tau$$

$$\frac{E_r}{\eta_1} \cos \theta_i - \frac{E_i}{\eta_1} \cos \theta_i = -\frac{E_t}{\eta_2} \cos \theta_t \rightarrow ②$$

Parallel polarization: TM



$$H_i + H_r = H_t$$

$$\frac{E_i}{\eta_1} + \frac{E_r}{\eta_1} = \frac{E_t}{\eta_2} \rightarrow (2)$$

$$= E_r \cos \theta_i + E_i \cos \theta_i = E_t \cos \theta_t \rightarrow (2)$$

$$\frac{E_r}{E_i} = \frac{\eta_1 \cos \theta_i - \eta_2 \cos \theta_t}{\eta_1 \cos \theta_i + \eta_2 \cos \theta_t}$$

$$\frac{E_t}{E_i} = \tau = \frac{2 \eta_2 \cos \theta_i}{\eta_1 \cos \theta_i + \eta_2 \cos \theta_t}$$

Special Cases:



Normal Incidence

Parallel polarisation

$$\left[\Gamma = \frac{\eta_1 - \eta_2}{\eta_1 + \eta_2} \right]$$

$$\tau = \frac{2\eta_2}{\eta_1 + \eta_2}$$

Perpendicular polarisation

$$\Gamma = \frac{\eta_2 - \eta_1}{\eta_2 + \eta_1}$$

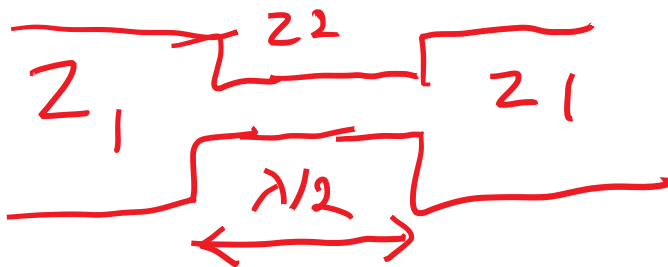
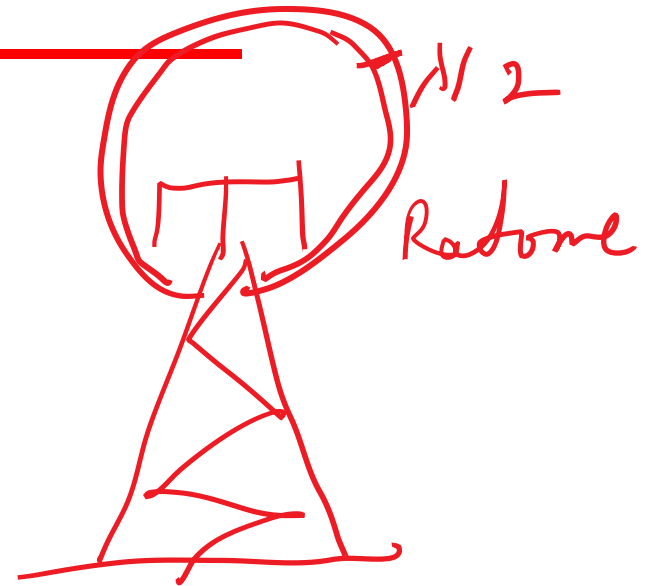
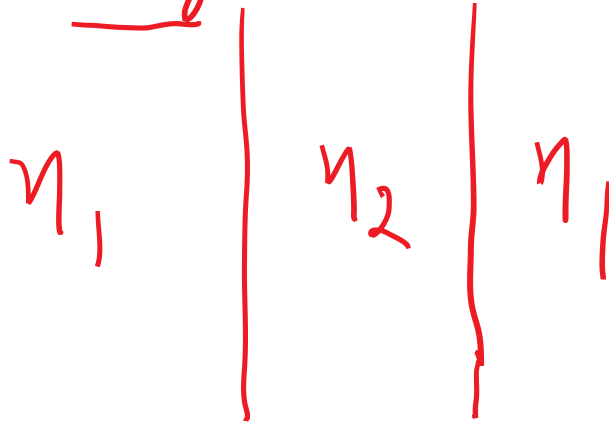
$$\tau = \frac{2\eta_2}{\eta_2 + \eta_1}$$

Special Cases:



$\lambda/2$

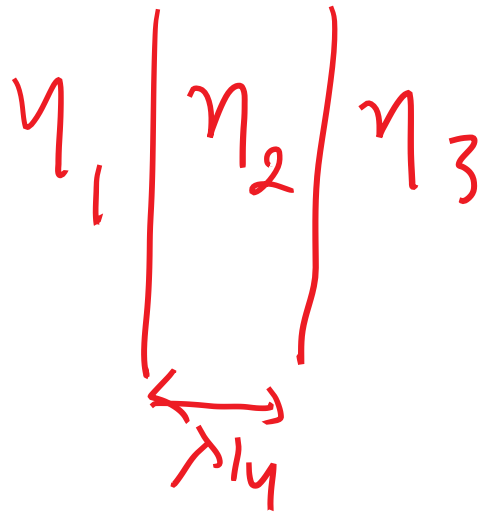
length



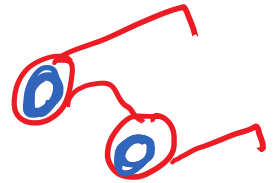
Special Cases:



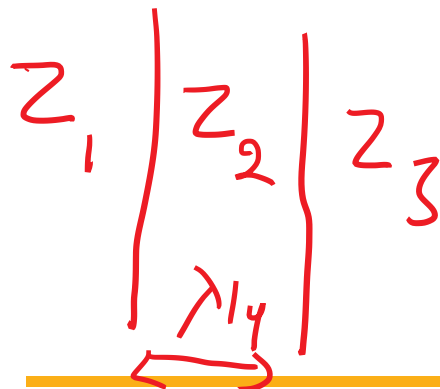
$\lambda/4$ length



Antireflection
coating

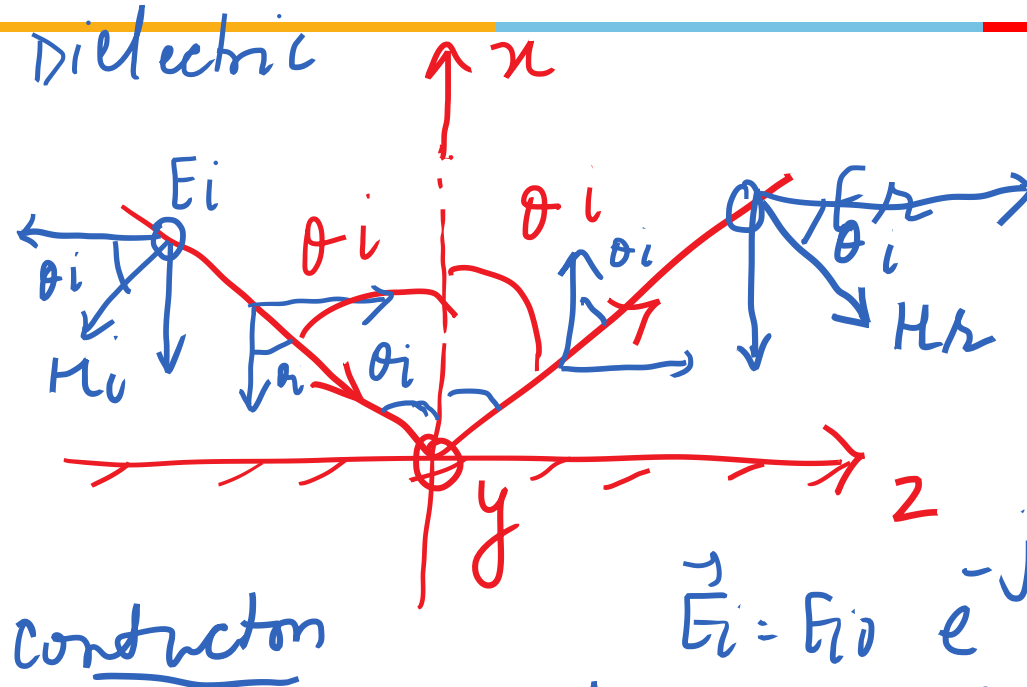


$$\underline{\eta_2 = \sqrt{\eta_1 \eta_3}}$$



$$Z_2 = \sqrt{Z_1 Z_3}$$

EM waves at dielectric conductor interface (TE Wave)



$$z=0$$

$$E_i + E_r = 0$$

$$\boxed{\Gamma = -1}$$

conductor

$$\vec{E}_i = E_{i0} e^{-j\beta[-\cos\theta_i z + \sin\theta_i x]} \hat{a}_y$$

$$\vec{E}_r = E_{r0} e^{-j\beta[\cos\theta_i z + \sin\theta_i x]} \hat{a}_y$$

$$(-\cos\theta_i \hat{a}_z - \sin\theta_i \hat{a}_x) e^{-j\beta[-\cos\theta_i z + \sin\theta_i x]}$$

$$\vec{H}_i = H_{i0}$$

$$\vec{H}_r = H_{r0} (-\sin\theta_i \hat{a}_x + \cos\theta_i \hat{a}_z) e^{-j\beta[\cos\theta_i z + \sin\theta_i x]}$$

EM waves at dielectric conductor interface (TE Wave)



$$E_{\text{Tot}} = E_i + E_r$$

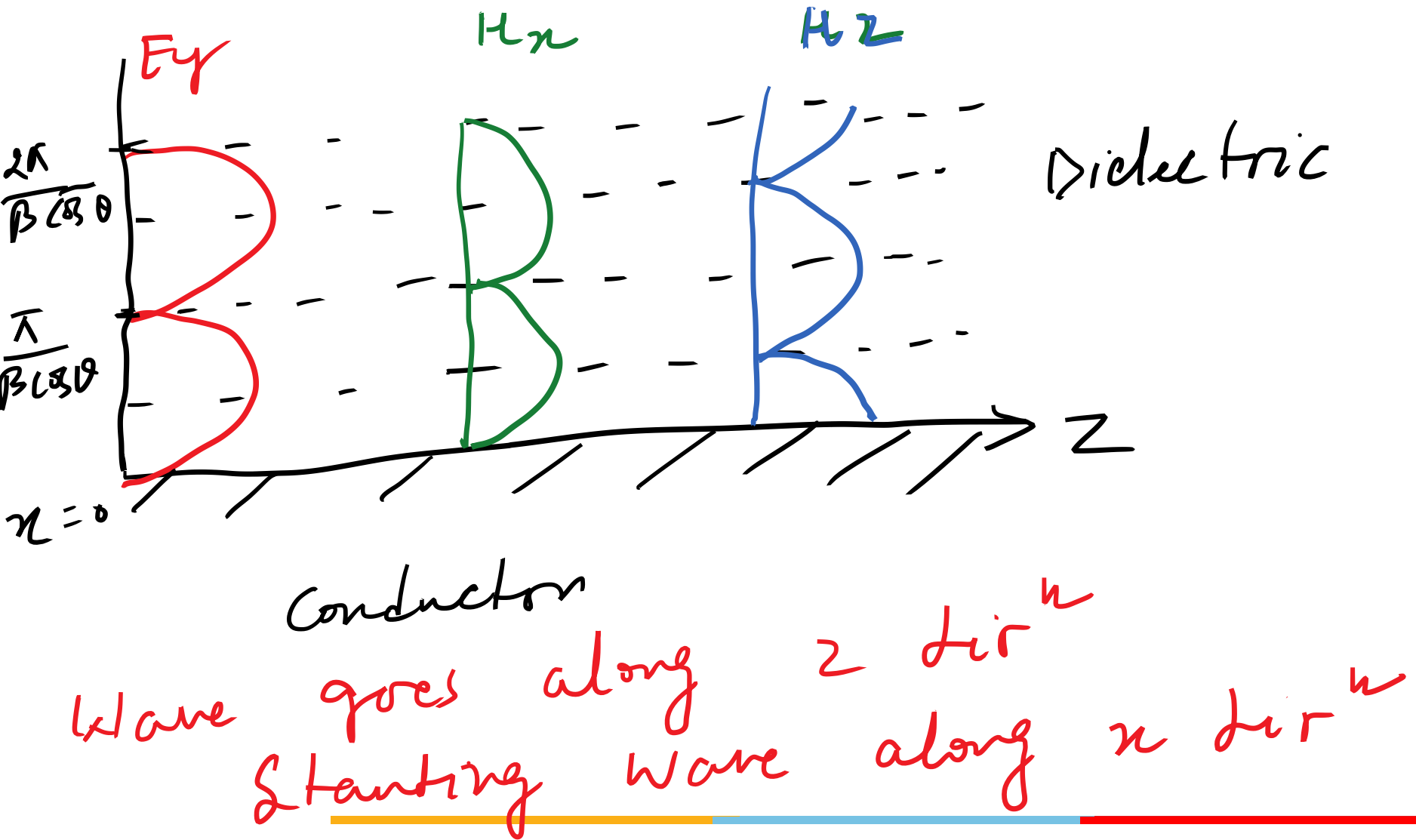
$$= E_{i0} e^{-j\beta(-x \cos \theta_i + z \sin \theta_i)} + E_{r0} e^{-j\beta(x \cos \theta_i + z \sin \theta_i)} \hat{a}_y$$

$$E_{\text{tot}} = E_{i0} e^{-j\beta z \sin \theta_i} \left[e^{j\beta x \cos \theta_i} - e^{-j\beta x \cos \theta_i} \right] \hat{a}_y$$

$$\left[E_{\text{tot}} = 2j E_{i0} e^{-j\beta z \sin \theta_i} \sin(\beta x \cos \theta_i) \hat{a}_y \right]$$

$$\underline{\underline{H_{\text{tot}}}} = -\frac{E_{i0}}{\eta} e^{-j\beta z \sin \theta_i} \left[2 \cos \theta_i \cos(\beta x \cos \theta_i) \hat{a}_z + 2j \sin \theta_i \sin(\beta x \cos \theta_i) \hat{a}_n \right]$$

EM waves at dielectric conductor interface (TE Wave)

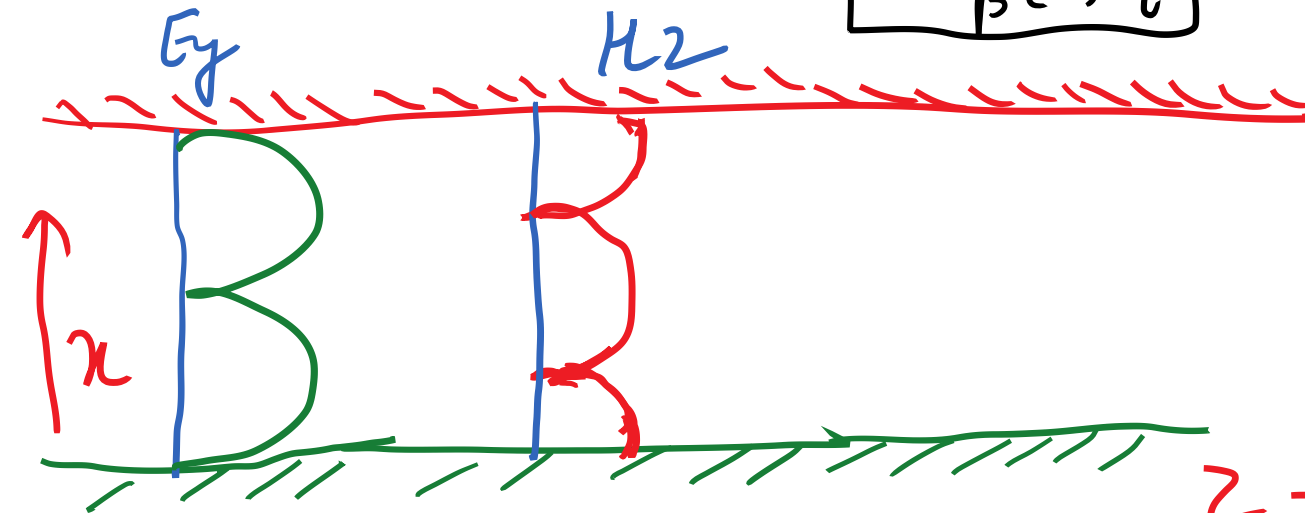


Parallel Plate Waveguide



$$m=1, 2, 3, \dots$$

$$\alpha = \frac{m\pi}{\beta \cos \theta}$$



Conductor

Placing an
another
conductor
on other
side.

So wave will be guiding between
two parallel plate.

Parallel Plate Waveguide



Condition for wave to propagate in parallel plate waveguide

$$\underline{d} = \lambda = \frac{m\lambda}{\beta \cos \theta_i}$$

$$\cos \theta_i = \frac{m\lambda}{\beta d} = \frac{m\lambda}{2d}$$

$$\underline{\cos \theta_i \leq 1}$$

$$d \geq m\lambda/2$$

($m=1$)

$$\boxed{d \geq \frac{\lambda}{2}}$$

